



Dark Forest Internet

The Internet has become more insular and closed with the growth of private channels. <https://digit.in/june19-64>



The Persistence of Chaos

Guo O Dong is selling a laptop filled with 6 deadly strains of malware for over \$1 million. <https://digit.in/june19-65>



In short supply

Agent001 has to help a friend navigate the cryptic world of power supply technologies, tiers and certifications.

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A

few days ago, I received a rather peculiar query from a friend. He'd wished to upgrade his computer

with a new graphics card and had gone through a few articles about power supplies to figure out what he'd need. He'd managed to calculate his power requirement but was confused regarding a few terms he'd come across. Apparently, having referred to multiple forums, he was introduced to the concept of PSU tiers. While certification such as the 80+ Bronze, Silver, Gold, Platinum, Titanium was easy to comprehend, the whole concept of tiers wherein PSUs of different certifications were present in the same tier was confusing. If two power supplies are indeed Gold certified, then why on earth are they in two different tiers.

Like him, many others had taken to the concept of tiers rather seriously. That's not a bad thing, it's just that you need a bit more context about the concept of tiers before you go through them. And most places that mention these tiers don't provide said context. Tiers are simply a way to rank the power supplies based on community feedback which is sometimes clubbed with empirical testing. Since the feedback element varies from one community to another, you will see tier lists with a few power supplies

classified higher or lower. Say if two or more members of a particular community were to say that a certain power supply burned out on them, then the PSU tier list formulated by that community will have that particular power supply downgraded by a tier. Which is why it's always better to rely on a tier list that has an empirical element to it. Very few PC review sites



Seasonic Focus Gold PSU

properly test power supplies because you need a lot of expensive equipment to properly test them. The rest simply rely on the websites that do empirical testing but as time passes, the lists start to differ a bit.

There are certain lists that you can always trust such as the infamous JonnyGURU tier list which has been trusted for many years. However, there's another factor at play here, the brands. When certain brands get good reviews for a PSU model, they'll try to capitalise on this by trying to retain

the same model number even after changing the power supply design. Some brands will have a version number placed in a nondescript location while some brands will try to cheat you in an outright fashion. This is where you need to cross check with the certification authority. Entities which offer certification such as Plug Load Solutions which offers the 80+ badge, and Underwriters Laboratories which offers the UL certification or TÜV SÜD, all document each and every power supply they test and provide unique file numbers for each new design. This way you can tell if the design has changed or not.

After learning the truth about PSU tiers and certifications, my friend finally decided that he didn't need a power supply with a ridiculous overhead and finally settled on a Seasonic Focus Gold which he managed to get for cheap from a friend travelling back from the States. An 80+ Silver certified power supply is actually proper for everyone. You should only bother with a Gold or higher level certified power supply if the price difference is very low. The law of diminishing returns applies heavily here. Power supply manufacturers have managed to nearly perfect PSU designs so there's very little difference between good PSUs that are manufactured by different companies. All you need to worry about is user feedback. Simply